

Tutorial 8

Linear independence, basis and dimension

1. *This question asks you to try to understand what linear dependence means geometrically, in $\mathbb{R}^3$. Try to visualize the sets of vectors.*

For each of the following sets of vectors in $\mathbb{R}^3$, say whether it is

- (i) always (ii) never (iii) sometimes
linearly independent.

If you choose (iii), say when the vectors will be linearly independent.

- (a) Two distinct (different) non-zero vectors on a plane through the origin.

Solution: Sometimes; when the one vector is not a multiple of the other.

- (b) Three distinct non-zero vectors on a plane through the origin.

Solution: Never.

- (c) Two distinct non-zero vectors in $\mathbb{R}^3$.

Solution: Sometimes; when the one vector is not a multiple of the other.

- (d) Three distinct non-zero vectors in $\mathbb{R}^3$.

Solution: Sometimes; when the vectors do not all lie in the same plane through the origin.

2. Let S and T be subsets of $M_{2 \times 2}(\mathbb{R})$ defined by

$$S = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} \right\} \text{ and } T = \left\{ \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} \right\}.$$

In Question 6 in Tutorial 7 you found that

$$\text{span } S = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(\mathbb{R}) : a + b - c - d = 0 \right\},$$

and that it is possible to find a 2×2 matrix which is not in $\text{span}(S)$.

- (a) Is S a linearly independent subset of $M_{2 \times 2}(\mathbb{R})$? Give a reason for your answer.

Solution: To check for linear independence, consider the equation

$$\lambda_1 \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

This leads to the system of linear equations

$$2\lambda_1 + 3\lambda_2 + \lambda_3 = 0$$

$$-\lambda_1 + 2\lambda_3 = 0$$

$$\lambda_2 + 3\lambda_3 = 0$$

$$\lambda_1 + 2\lambda_2 = 0$$

Putting $a = b = c = d = 0$ in the augmented matrix in Question 6(a) of Tutorial 7, we see that the only solution is $\lambda_1 = \lambda_2 = \lambda_3 = 0$, so S is linearly independent.

Try to use your work in Question 6 of Tutorial 7 to answer the following questions quickly.

- (b) Is T a linearly independent subset of $M_{2 \times 2}(\mathbb{R})$? Give a reason for your answer; you should be able to do this without doing any calculations.

Solution: Certainly T is linearly independent, because neither matrix in T is a scalar multiple of the other. Alternatively, you could use the fact that any subset of a linearly independent set is still linearly independent. (Make sure you can prove this.)

- (c) Find $\text{span } T$. How many equations will be needed to describe $\text{span } T$? Find the equation or equations. Write your answer in the same form as that for $\text{span}(S)$ above. You should be able to use your working for Question 6 of Tutorial 7.

Solution: Look at the working where you found the span of S . Instead of considering the equation

$$\lambda_1 \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 2 \\ 3 & 0 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

you will now consider

$$\lambda_1 \begin{pmatrix} 2 & -1 \\ 0 & 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}.$$

So you can use your previous working (just eliminating the column for λ_3). Two equations describing the span of T are for instance $b + d - 2c = 0$ and $a + b - c - d = 0$.

3. Without doing any lengthy calculations, state whether or not the following subsets of vectors form a basis for the given vector space. Give reasons for your answers. In this question you can use any of the useful facts we have covered in lectures concerning bases. This should go quickly; use your notes!

(a) $\left\{ \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 1 & 2 \\ 5 & 6 \end{pmatrix}, \begin{pmatrix} 9 & 2 \\ 1 & 4 \end{pmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{R})$.

Solution: The given set is not a basis for the space: $M_{2 \times 2}(\mathbb{R})$ has dimension 4, and the given set has only 3 vectors, so it does not span $M_{2 \times 2}(\mathbb{R})$.

(b) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$ in $\mathbb{R}^3$.

Solution: The given set is not a basis for the space: it is linearly dependent because it contains the zero vector.

(c) $\{x^2 + x, 3x, 4x^2 + 1, -17\}$ in P_2 .

Solution: The set is not a basis, since P_2 has dimension 3, but the given set has 4 elements and therefore has to be linearly dependent.

(d) $\{x, x - 1, x^2 - x\}$ in P_2 .

Solution: The given set is a basis for the space: No vector in the given set is a linear combination of the other two vectors in the set (you can see this by looking at the powers of x), so the given set is independent. The given set has 3 vectors in it, and P_2 has dimension 3, so the given set is a basis for P_2 . (Use Theorem 2.5.21.)

(e) $\left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 4 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\}$ in $\mathbb{R}^4$.

Solution: The given set is not a basis for the space: The given set is dependent, because its second vector is a scalar multiple of the first.

4. Let $B = \{x^2 + 1, x - 1, x^2 - x\}$.
 (a) Show that B is a basis for P_2 .

Solution: To show that $\text{span } B = P_2$, we need to show that for every choice of $a, b, c \in \mathbb{R}$, we can find real numbers $\lambda_1, \lambda_2, \lambda_3$ such that

$$\lambda_1(x^2 + 1) + \lambda_2(x - 1) + \lambda_3(x^2 - x) = a + bx + cx^2 \quad (*)$$

This can be written as

$$(\lambda_1 - \lambda_2) + (\lambda_2 - \lambda_3)x + (\lambda_1 + \lambda_3)x^2 = a + bx + cx^2.$$

Equating coefficients of powers of x on both sides, we obtain a system of linear equations with augmented matrix

$$\begin{pmatrix} 1 & -1 & 0 & a \\ 0 & 1 & -1 & b \\ 1 & 0 & 1 & c \end{pmatrix} \text{ which reduces to } \begin{pmatrix} 1 & -1 & 0 & a \\ 0 & 1 & -1 & b \\ 0 & 0 & 2 & c - a - b \end{pmatrix}. \quad (**)$$

This system is consistent (has a solution) for all values of a, b, c , and therefore $\text{span } B = P_2$. To check whether B is linearly independent, take $a = b = c = 0$ in $(*)$. This leads to a homogeneous system with augmented matrix

$$\begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 1 & 0 & 1 & 0 \end{pmatrix} \text{ which reduces to } \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 0 \end{pmatrix}.$$

This system has only the trivial solution, showing that B is linearly independent, and hence a basis.

- (b) Find the coordinate vector of $5x^2 - 9$ with respect to the basis B .

Solution: To get the coordinate vector of $p(x) = 5x^2 - 9$, we have to solve the equation $(*)$ with $a = -9, b = 0, c = 5$. It follows from $(**)$ that $\lambda_1 = -2, \lambda_2 = 7, \lambda_3 = 7$, and so the coordinate vector of $p(x)$ is $(-2, 7, 7)$.

5. Read the following result and its proof, and then answer the questions following it.

Theorem Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a linearly independent set in a vector space V . If $\mathbf{v}$ is a vector in V that is not in the span of S then $S \cup \{\mathbf{v}\}$ is still linearly independent.

Proof. Suppose that $\mathbf{v} \notin \text{span}(S)$, but $S \cup \{\mathbf{v}\}$ is linearly dependent. Then there are scalars $\lambda_1, \lambda_2, \dots, \lambda_{n+1}$, not all zero, such that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n + \lambda_{n+1} \mathbf{v} = \mathbf{0}.$$

We have $\lambda_{n+1} \neq 0$, since S is linearly independent. Then

$$\mathbf{v} = -\frac{1}{\lambda_{n+1}}[\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_n \mathbf{v}_n].$$

But this is a contradiction, so $S \cup \{\mathbf{v}\}$ must be linearly independent. □

- (a) What method of proof is used to prove this result?

Solution: Proof by contradiction.

- (b) How does it follow from the linear independence of S that $\lambda_{n+1} \neq 0$?

Solution: If $\lambda_{n+1} = 0$, then there would be an i with $1 \leq i \leq n$ such that $\lambda_i \neq 0$ and $\lambda_1 \mathbf{u}_1 + \dots + \lambda_n \mathbf{u}_n = \mathbf{0}$, contradicting the linear independence of S .

- (c) What is the contradiction referred to in the last line of the proof?

Solution: The second equation in the proof shows that $\mathbf{v} \in \text{span}(S)$, contradicting the assumption that $\mathbf{v} \notin \text{span}(S)$.

- (d) Let S be the set defined in Question 2. Is there a set B **containing** S which has four matrices in it and is independent? If so, find one; if not, explain why not. (If you have understood the previous questions, this should require no calculation at all.)

Solution: We can take $B = S \cup \{A\}$ where A is any 2×2 matrix not in $\text{span } S$, e.g. $A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. The theorem proved here shows that B will then be linearly independent.

- (e) Will the theorem still be true if we replace “If $\mathbf{v}$ is a vector in V that is not in the span of S then ...” by “If $\mathbf{v}$ is a vector in V that is not in S then ...”? Give a reason for your answer.

Solution: The theorem will no longer be true. For example, use $\mathbb{R}^3$ for V ; let $S = \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$ and $\mathbf{u} = \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix}$. Then S is linearly independent and $\mathbf{u} \notin S$ but $S \cup \{\mathbf{u}\}$ is linearly dependent.